MissionMind — Project Audit & Research Direction

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Summary

MissionMind is not yet a research result. The ingestion code runs, but the query layer and the evaluation harness cannot run. No experiment has been executed or recorded, and the core idea, GraphRAG over failure reports, is already published in several domains. A defensible IEEE contribution is still realistic. It needs a sharper research question, a larger and correctly sourced corpus, human-labelled evaluation data, and fair baselines.

Most consequential findings (all verified on 2026-10-12 against commit `5d0219e`):

- **Measured.** The NER-first extraction passes its own gate on 13 of 2,175 chunks (0.6%). The other 99.4% are routed to the Llama-3 fallback, so the knowledge graph would be almost entirely unvalidated LLM output.
- **Verified.** `src/graph/retriever.py` holds a FastAPI app that imports `GraphRAGRetriever` from itself. That class no longer exists anywhere in the repository, so the API and the CLI both fail at import.
- **Verified.** `evalutation/run_eval.py` imports `extract_failure_data`, which does not exist. `test_data.json` is still a placeholder template. There are no results.
- **Verified.** The corpus contains no ISRO failure report, and the promised Chandrayaan-2 and Mars Polar Lander reports are missing. The Chandrayaan-2 Failure Analysis Committee report has never been made public. One of the two ESA documents is a 2018 journal article, not the inquiry report.
- **Verified.** The novelty claim needs repositioning. GraphRAG or KG-RAG over failure records already exists for automotive (IEEE BigData 2024), FMEA (J. Ind. Inf. Integration 2025), aviation (IEEE ICSC 2026) and maritime (IEEE CASE 2026) data.

Recommended direction. Reframe the work as *evidence-anchored, cross-mission causal analogy retrieval over long, heterogeneous spaceflight mishap reports*, using the IEEE CASE 2026 maritime RCA paper as the base. That paper names graph-based reasoning as its future work and lists proxy relevance labels and an unvalidated LLM judge as its limitations. Our project can address all three on a domain the paper does not cover. The revised proposal sets out the design. Implementation waits for your approval.

1. Audit method and evidence base

Every claim below comes from reading the code at commit `5d0219e`, inspecting the data on disk, or running offline checks. Claims from the proposal or README were not taken at face value.

What	How it was checked	Executed?
Proposal (`MissionMind_Proposal (2).docx`, 7 sections, ~600 words)	Full text extracted and read	Yes
All 15 Python modules, README, requirements, `.gitignore`, git history (7 commits)	Read line by line	Yes
Corpus: 7 PDFs (128 MB), 7 extracted texts, 7 chunk files	Word and chunk counts, opening pages, corruption rate	Yes
NER layer (`ner_model.py`) and the gate in `graph_builder.py`	Ran `FailureNER` over all 2,175 chunks offline (324 s) and probed case sensitivity	Yes
Neo4j ingestion, graph build, LLM fallback, API, evaluation	Static analysis only. Neo4j and Ollama were not running, and the code cannot reach the query layer anyway (Section 3)	No
Environment	Python 3.14.2; all imports present; RTX 4060 Laptop GPU (8 GB), CUDA available	Yes

Limitation of this audit: no graph-build run was executed, so graph-level statistics (node counts, edge counts, LLM extraction quality) are not reported. No logs or outputs from an earlier run are committed, so none could be inspected either.

2. Claims against evidence

Of 12 claims checked, 1 is supported, 6 are partially supported, 3 are unsupported and 2 are contradicted by the code or data.

Claim (source)	Evidence in the repository	Verdict
Curated corpus of NASA, ISRO and ESA investigation reports, "approximately 15–25 cases" (proposal §3)	5 failure documents: 3 NASA, 2 ESA. The ISRO material is 2 annual reports, not failure reports. Chandrayaan-2 and Mars Polar Lander are absent	Contradicted
NER extracts subsystem, mission phase, failure type and root-cause category (proposal §4)	Gazetteer of 23 + 19 + 20 terms, case-sensitive. `mission_phase` is extracted but never written to the graph (`graph_builder.py:162-177`)	Partial
"Transformer-backed NLP pipelines" for FMEA NER (README)	`en_core_web_sm` (a CNN pipeline) plus an `EntityRuler`. The proposal says BERT; the embedder is MiniLM-L6, a distilled BERT-family model	Partial
BERT embeddings capture similarity between failures (proposal §4)	`all-MiniLM-L6-v2` similarity at an uncalibrated 0.65 threshold. The code itself warns not to report it unjustified (`embeddings.py:24, 104-112`)	Partial
KG with nodes for missions, subsystems, root causes and similarity edges (proposal §4)	Cypher is written (`graph_builder.py:162-200`) but never verified by a run. `FailureMode` and `RootCause` nodes are merged by raw string	Partial
Hybrid Cypher + vector retrieval using `neo4j-graphrag` (README)	`neo4j-graphrag` is never imported. The retriever class was deleted (Section 3)	Unsupported
"Deterministic, hallucination-free" LLM querying (README)	99.4% of graph facts would come from LLM extraction with no grounding check, and generation is unconstrained	Contradicted
"Cross-agency causal discovery … linking an ISRO telemetry anomaly to an ESA flight software failure" (README)	No ISRO failure data exists. No cross-mission output has been produced or evaluated	Unsupported
RAGAS faithfulness and context precision evaluation (README)	`ragas` is in requirements but never imported. The custom harness is broken	Unsupported

Claim (source)	Evidence in the repository	Verdict
Source-grounded, cited explanations (proposal §5, §7)	Chunks carry no page numbers, and whitespace collapse destroys section structure. Answers cannot cite a page	Partial
Local Llama-3 synthesis via Ollama (README)	`ollama.chat(model="llama3")` is present in 4 modules	Supported
Interactive demo with graph visualisation (proposal §7)	No frontend or visualisation code exists. The API cannot import	Partial (planned only)

3. Critical defects in the implementation

Three defects block any experiment. Six more would invalidate results even after the blockers are fixed.

3.1 Blockers

1. **Query layer cannot import.** `src/graph/retriever.py` contains the FastAPI app (its docstring even says `main.py`). Line 17 imports `GraphRAGRetriever` from `graph.retriever`, i.e. from itself. The class existed in commit `7d9ce24` and was overwritten in `5d0219e`. Neither `build_context_text()` nor the `related_failures` field used by the API exists anywhere. So the cross-mission context promised in the prompt is never assembled.
2. **Evaluation cannot run.** `evalutation/run_eval.py:30` imports `extract_failure_data` from `graph_builder`. No such function exists. `test_data.json` holds 2 placeholder chunks with `"<paste the actual chunk text…>"` and 1 query.
3. **No recorded run.** There is no `results.json`, no graph export and no log. Nothing in the repository shows that the four-step pipeline has completed end to end.

3.2 Extraction is LLM-only in practice (measured)

The NER path is used only when a chunk has a gazetteer entity **and** a root-cause keyword (`graph_builder.py:137`). Running the project's own `FailureNER` over every chunk gave:

Document	Chunks	Has entity	Has root-cause keyword	Passes NER gate
NASA Columbia CAIB Vol. 1	874	76	89	6

Document	Chunks	Has entity	Has root-cause keyword	Passes NER gate
Challenger (Rogers Commission)	596	46	136	4
ISRO Annual Report 2025–26	300	10	3	0
ISRO Annual Report 2024–25	244	8	5	0
Mars Climate Orbiter MIB Phase I	65	18	2	0
ExoMars Schiaparelli (journal article)	65	5	2	0
Ariane 5 Flight 501 Inquiry Board	31	12	4	3
Total	**2,175**	**175**	**241**	**13 (0.6%)**

Consequences: 2,162 chunks each need a Llama-3 call. For a sense of scale only, that is roughly 3–9 hours at an assumed 5–15 s per call on the RTX 4060 (not measured). Each graph fact is then accepted with no check that it appears in the source text. Mars Climate Orbiter, whose root cause is a unit mismatch, gets no NER-path extraction at all.

3.3 Defects that would corrupt results

1. **Case-sensitive gazetteer.** `EntityRuler` string patterns match exact token text. A probe showed "The Solid Rocket Booster failed" yields no entity while the lowercase form does. Report text is often capitalised, so recall is depressed.
2. **Artifactual cross-mission links.** `FailureMode` nodes are `MERGE`d by name across missions. The fallback name `"<category> related failure"` is shared. On this corpus, `"material failure related failure"` is produced for both Challenger and Columbia, which creates a cross-mission connection out of a template string.
3. **First-match root-cause labelling.** `categorize_root_cause` returns the first dictionary category with any keyword hit. Dictionary order therefore biases labels, multi-causal chunks get one label, and keywords like `"erosion"` or `"oversight"` fire on non-causal uses.
4. **One record per (mission, failure mode) for similarity.** `graph_builder.py:179-184` keeps only the first root cause seen. Pairwise similarity then runs on arbitrary representatives at an uncalibrated threshold.
5. **`mission_phase` discarded.** One of the four structured fields in the proposal's objective never reaches the graph.

6. **Same model as generator and judge.** `JUDGE_MODEL = GENERATION_MODEL = "llama3"` invites self-preference bias. Field matching uses a fixed MiniLM 0.6 cosine threshold that has never been validated.

4. Corpus and data quality

The corpus cannot support the project's central claim. It covers 5 failures from 2 agencies, so there are at most 10 cross-mission pairs. Two of its seven documents (25% of chunks) are not about failures.

Document on disk	What it actually is	Words	Issue
`NASA_Columbia_CAIB_Vol1`	CAIB Report Vol. I (2003)	174,712	40% of all chunks; dominates retrieval
`Challenger`	Rogers Commission Report Vol. 1 (1986), scanned	119,088	Heavy OCR noise ("CEALLEMGEB", "Sfiace") that the quality scorer passed
`Annual_Report_2025_26_Eng…`	ISRO / Dept. of Space annual report	60,013	Not a failure report
`Annual_Report_2024_25_Eng`	ISRO / Dept. of Space annual report	48,679	Not a failure report
`Mars_Climate_Orbiter`	MCO Mishap Investigation Board Phase I (1999)	12,993	Correct source
`ExoMars_2016_Schiaparelli…`	Aboudan et al., *Space Sci. Rev.* 214:97 (2018), a trajectory reconstruction paper	12,856	Not the ESA inquiry report. Springer copyright limits redistribution
`Ariane_5_Flight_501…`	Ariane 5 Flight 501 Inquiry Board Report (1996)	6,076	Correct source

Text-processing defects found:

- **Ligatures destroyed.** `sanitize_text` deletes every non-ASCII character before any normalisation. "fi"/"fl" ligatures vanish, giving "Pro les" for *Profiles* and "ew" for *flew*. Unicode NFKC normalisation should run before any filtering.

- **Structure lost.** All whitespace is collapsed before chunking. Headings, sections, findings and recommendations become indistinguishable, and fixed 250-word windows cut across them.
- **No provenance below document level.** Page numbers are discarded, so no answer can cite a page and no extraction can be audited against its source span.
- **Not reproducible.** `data/` is git-ignored, and there is no manifest of source URLs, versions or checksums.

Source availability, checked on 2026-10-12:

- The ESA Schiaparelli Anomaly Inquiry report (DG-I/2017/546/TTN, May 2017) is public and should replace the journal article.
- ISRO published the GSLV-F10 Failure Analysis Committee conclusions as a short web statement, which is usable but thin.
- The Chandrayaan-2 FAC report has not been made public according to The Wire. The proposal's headline example, "Apollo 13 and Chandrayaan-2", cannot be grounded in an official source.
- Balanced ISRO coverage is therefore not achievable from public documents. This is a scope limitation to state openly, not to paper over.

5. Evaluation, reproducibility and engineering hygiene

Even if the harness ran, it could not produce publishable evidence. There is no human-labelled data, no baseline beyond "chunks without graph facts", no significance testing, and the judge is the generator.

Evaluation design gaps

- There is no gold standard. Extraction, retrieval and answer quality all need human labels with a written annotation guideline and inter-annotator agreement.
- `extraction_metrics` scores only the LLM path. Field correctness uses a single cosine threshold that has never been validated against human judgement.
- Retrieval is scored at k = 3 only, on chunk IDs. Cross-mission questions need mission-level and passage-level relevance.
- The only ablation is graph facts versus none. There is no BM25, hybrid, LLM-only, or published GraphRAG baseline (e.g. Microsoft GraphRAG, LightRAG).
- There are no question categories, so the design cannot show *where* a graph helps. Aggregation and cross-mission questions are where graph structure is expected to matter.
- There are no repeated runs, confidence intervals, or paired significance tests.

Reproducibility and hygiene

Issue	Location	Fix
Unpinned dependencies; `ollama`, `wordfreq`, `pytesseract`, `pdf2image` missing from requirements	`requirements.txt`	Pin versions and add a lock file
Hard-coded credentials and default password `"password"`	`src/backend/main.py:14-16`, `run_eval.py:36-38`, connector default	Environment only, no fallback secret
Duplicate and dead code: empty `backend/`, legacy `src/backend/main.py`	repo root, `src/backend/`	Remove or consolidate
Folder typo `evalutation/` (README says `evaluation/`)	repo root	Rename
`sys.path.append` imports instead of a package	every module	`pyproject.toml` and an installable package
No tests, no config file, no seeds, no model or prompt versioning	—	Config-driven runs, logged prompts, fixed seeds
Graph writes one transaction per chunk; all chunks are re-embedded even when skipped	`neo4j_loader.py:57-88`	Batch `UNWIND` writes; embed only new chunks

What works and should be kept: PyMuPDF extraction with page-level OCR fallback; the embedded-digit corruption heuristic; idempotent, skip-if-done steps; the shared Neo4j connector; and the honest in-code caveats (e.g. the threshold-calibration note).

6. Novelty and IEEE sufficiency

As it stands, the contribution is insufficient for a reputable IEEE venue. The pipeline (PDF → chunks → NER/LLM triples → Neo4j → vector + graph context → LLM) is the standard KG-RAG recipe. The domain application alone is not new, and there is no evidence that the system works.

What is genuinely under-served, and therefore a credible gap:

- **Spaceflight mishap reports specifically.** Prior KG/RAG work targets aviation (NTSB, ASRS), maritime tribunals, manufacturing FMEA and automotive records. These are large collections of short, semi-structured, single-format records. Spaceflight investigation reports are few (tens), very long (6k–175k words), unstructured, and heterogeneous across agencies and decades. Methods that rely on standard fields or metadata, as the CASE 2026 base paper does, do not transfer directly.
- **Cross-mission causal analogy as a retrieval task.** "Which other failures share this causal pattern?" is the analyst's real question. None of the papers reviewed evaluates it against human analogy judgements. Human analyses of cross-mission failure patterns exist (e.g. Tafazoli 2009, on-orbit failures by subsystem), but they are not retrieval benchmarks.
- **Evidence-anchored graph construction.** Most KG-RAG pipelines accept LLM-extracted triples unverified. AGRAG (IEEE ICDE 2026) names LLM hallucination in graph construction as a core weakness of GraphRAG. Requiring every causal edge to carry a verifiable source span and page is a concrete, testable remedy.

The minimum for a credible IEEE conference paper is a precise research question, a reusable labelled resource, a method justified against strong baselines, and statistically supported results with limitations. The revised proposal is built around exactly these four. A journal (e.g. IEEE Access) would additionally expect larger scale or a second domain to show generalisation.

7. Base paper: search and selection

Selected: S. Kim and S. Kim, "Multi-Field Hybrid Retrieval-Augmented Generation for Maritime Accident Root Cause Analysis," *2026 IEEE 22nd International Conference on Automation Science and Engineering (CASE)*, Aug. 2026, pp. 238–243, doi: 10.1109/CASE69030.2026.11704480. Full text: arXiv:2606.13249 (CC BY 4.0).

How status was verified. The DOI resolves (HTTP 302) to IEEE Xplore document 11704480. Crossref citation metadata gives the venue as the 2026 IEEE 22nd CASE, pp. 238–243, Aug. 2026. OpenAlex links the IEEE DOI to arXiv 2606.13249, confirming they are the same work, and all 6 pages of the arXiv version were read. Crossref lists the surname "Kim" three times while arXiv lists two authors (Seongjin Kim, Sungil Kim); cite the two authors as printed on the paper.

Search method. Web searches, then OpenAlex queries for 2026 works with IEEE DOIs (prefix 10.1109) on 10 topic strings. Crossref direct search was rate-limited (HTTP 429). IEEE Xplore pages did not render for automated access, so venue facts rest on DOI and Crossref metadata.

Candidate (2026, IEEE DOI verified)	Relevance	Full text read?	Decision
Kim & Kim, maritime RCA multi-field hybrid RAG, IEEE CASE 2026	Investigation reports → precedent retrieval → grounded RCA; names graph reasoning as future work	Yes (arXiv)	**Base paper**
Sanyaolu & Josyula, KG construction from NTSB narratives, IEEE ICSC 2026, pp. 397–402, doi	Schema-constrained LLM KG over 6,193 accident reports; F1 0.87 extraction	Abstract only (paywalled)	Primary supporting paper
Wang et al., AGRAG, IEEE ICDE 2026, doi	General GraphRAG; statistics-based graph construction to avoid LLM hallucination	Abstract	Method reference and baseline candidate
Liang et al., GraphRAG Under Fire, IEEE S&P 2026, doi	Poisoning of GraphRAG via shared relations	Abstract	Threats to validity only
Tran et al., LLM-augmented KB for RCA, IEEE Access 2026, doi	Telecom support tickets	Abstract	Not selected (different data type)
Qiu et al., FMECA KG with LLM, IEEE RAITS 2026, doi	FMECA KG	No abstract available	Not selected (unverifiable content)

Not selected because the IEEE status could not be confirmed: arXiv 2604.13101 (aviation GraphRAG, "initial version of a conference publication", venue unnamed) and arXiv 2608.11277 (automotive HiL RCA). The latter was accepted at ICSRS 2026, but that conference is IEEE Reliability Society *co-sponsored*, and IEEE Xplore indexing of its proceedings could not be confirmed.

Why CASE 2026 over ICSC 2026. Both fit. ICSC 2026 is closer in graph construction but could not be read in full, so its limitations and future work cannot be cited accurately. The CASE paper is fully readable and states its limitations and future work explicitly. It also defines a concrete, reproducible baseline (multi-field BM25 + bge-m3 + RRF) that we can reimplement and compare against fairly.

8. Base paper analysis (Kim & Kim, IEEE CASE 2026)

The paper shows that field-aware hybrid retrieval over 13,329 tribunal reports raises proxy NormRecall@100 from 0.18 to 0.55. It also shows that grounding the generator lifts an LLM-judge score from 3.34 to 3.72. Every evaluation signal, however, is a proxy: metadata relevance, rule-based tags and an unvalidated LLM judge.

Aspect	What the paper does
Problem	Precedent search and RCA drafting over decades of Korea Maritime Safety Tribunal (KMST) adjudications is manual. Monolithic-text RAG dilutes field-specific signals
Data	13,329 KMST reports (1971–2025) → "incident cards" with Summary, Causes and Disposition fields → 37,007 chunks (mean 283 characters). 70/15/15 train/validation/test split
Labels	KMST L1/L2 cause taxonomy (24 L2 factors), assigned by rules: about 2,400 dictionary entries and 800 regexes, 97.6% coverage. No manual annotation
Retrieval	Per field: BM25 + dense (BAAI bge-m3). Six rankings fused by Reciprocal Rank Fusion, k_rrf = 60
Generation	Qwen3-Next-80B-A3B-Instruct, temperature 0, JSON schema. Outputs a cause narrative plus L1/L2 tags. Top-40 chunks (10–13 cases) as context
Retrieval metrics	Proxy RelScore = 0.5 keyword IDF-Jaccard + 0.4 accident type + 0.05 jurisdiction + 0.05 year proximity. Pseudo-gold = RelScore > 0.5. NormRecall@100 and nDCG@100
Generation metrics	ROUGE-L, SBERT cosine and LLM-judge (1–5) against the expert Causes field. L1 accuracy, L2 micro-F1 and Jaccard
Key results	NormRecall 0.1819 (whole-doc dense) → 0.5463 (multi-field hybrid). nDCG 0.6123 → 0.7697. Judge 3.340 → 3.723. SBERT 0.723 → 0.724. L1 0.771 → 0.774. L2 F1 0.460 → 0.490

Aspect	What the paper does
Stated limitations	(1) metadata proxy instead of expert relevance labels; (2) rule-based tags, not manual annotation; (3) LLM-as-judge model bias; (4) uncertainty-awareness of the evaluation needed
Stated future work	"integration of formal legal regulations, such as COLREGs, and graph-based reasoning to further expand the system's diagnostic depth and interpretability"

Reviewer-level observations on the base paper (relevant because our work must not repeat them):

- The paper says retrieval "significantly outperforms" baselines, but no significance test, confidence interval or repeated run is reported.
- Generation gains are small on every reference-based metric (SBERT +0.001, L1 +0.003). The headline +11.5% rests on the unvalidated judge. The judge model is not named.
- Weights (0.5, 0.4, 0.05, 0.05) were "empirically set", and the proxy is never checked against even a small human-labelled sample.
- The bge-m3 reference cites arXiv 2409.10216, whereas the M3-Embedding paper is arXiv 2402.03216 (ACL Findings 2024). This appears to be a citation error. It is noted only so we cite bge-m3 correctly ourselves.

Gaps our project can realistically address:

1. *Graph-based reasoning* (their future work) over precedents, implemented as typed causal-factor nodes shared across cases.
2. *Human relevance labels*, small but real, plus a meta-evaluation of how well a metadata proxy agrees with them (their limitation 1).
3. *Human-validated tags and judge*: agreement between the LLM judge and human ratings (their limitations 2 and 3).
4. *Transfer to long, unstructured, heterogeneous reports* where KMST-style fields do not exist and must be recovered by section segmentation.
5. *Abstention on unanswerable questions* as a first step toward their stated uncertainty-awareness.

9. Supporting literature and positioning

No reviewed work combines long, heterogeneous spaceflight mishap reports, evidence-anchored causal graphs, and human-labelled cross-case analogy evaluation. Each ingredient exists separately, so the contribution is in the combination and in the evaluation, not in any single component.

Work	Domain and data	Graph	Cross-case retrieval	Human-labelled evaluation	How we use it
Kim & Kim, IEEE CASE 2026	Maritime tribunal, 13,329 short cards	No	Yes (precedents)	No (proxy)	Base paper; multi-field hybrid RRF = baseline B3
Sanyaolu & Josyula, IEEE ICSC 2026	Aviation NTSB, 6,193 reports	Yes (schema-guided LLM KG)	Not reported in abstract	Extraction F1 on annotated sample	Schema-constrained extraction design
Ojima et al., IEEE BigData 2024, doi	Automotive failure records	Yes	Not verified	Not verified	Prior GraphRAG-for-failure evidence
Bahr et al., *J. Ind. Inf. Integr.* 45 (2025), doi	Manufacturing FMEA	Yes (FMEA ontology)	No	Retrieval recall/precision + human study	FMEA schema ideas
Edge et al., GraphRAG, arXiv 2404.16130	General corpora	Yes (communities)	Global summarisation	LLM-judged	Baseline B5 (global questions)
Guo et al., LightRAG, EMNLP Findings 2025, doi	General	Yes	Local + global	LLM-judged	Baseline B5 alternative (cheaper)
Wang et al., AGRAG, IEEE ICDE 2026	General	Yes (statistics-based)	Path reasoning	Benchmarks	Motivates verified graph construction
Tafazoli, *Acta Astronautica* 64(2–3), 2009, doi	On-orbit spacecraft failures	No (manual statistics)	Manual pattern analysis	Expert analysis	Domain motivation; subsystem taxonomy

Work	Domain and data	Graph	Cross-case retrieval	Human-labelled evaluation	How we use it
Tikayat Ray et al., aeroBERT-NER, AIAA SciTech 2023, doi; Andrade & Walsh, SafeAeroBERT, AIAA AVIATION 2023, doi	Aerospace text	No	No	Yes	Domain-encoder baselines for extraction
MissionMind (current)	5 space failures + 2 annual reports	Yes (unverified)	Intended, not implemented	None	—
MissionMind (proposed)	25–35 public mishap reports, multiple agencies (target)	Yes (evidence-anchored causal)	Yes (cross-mission analogy)	Yes (extraction, relevance, judge)	—

Cells marked "not verified" are where only the abstract or a search summary was seen. They must be checked against the full text before submission.

Evaluation methodology to adopt, all verified: RAGAS (Es et al., EACL 2024 demo, doi); ARES (Saad-Falcon et al., NAACL 2024, doi); FActScore atomic-claim factuality (Min et al., EMNLP 2023, doi); LLM-as-judge biases (Zheng et al., NeurIPS 2023, arXiv); paired bootstrap significance (Koehn, EMNLP 2004); position effects in long context (Liu et al., TACL 2024, doi).

10. Status of the work

Only ingestion is implemented and verified. Everything that produces a research claim is either unvalidated or not yet built.

1. Implemented and verified (in this audit)

- PDF text extraction for 7 documents, with page-level OCR fallback and a corruption heuristic. Output exists on disk.
- Word-window chunking (250 words, 50 overlap): 2,175 chunks on disk.
- `FailureNER` loads and runs. Its gate behaviour was measured (Section 3.2).

2. Implemented but not validated

- Neo4j chunk ingestion and the vector index (code reviewed, not run here).

- Knowledge-graph construction: NER path, LLM fallback, `MERGE` schema, similarity edges.
- Cross-mission similarity at the 0.65 threshold, and the `inspect_similarity_distribution.py` diagnostic.
- Metric functions in `metrics.py`. They are logically sound as pure functions but have never been executed on real data.

3. Broken (must be repaired, not just validated)

- Retriever, API and CLI (missing `GraphRAGRetriever`).
- Evaluation runner (missing `extract_failure_data`; placeholder test set).

4. Proposed for the next stages (detailed in the revised proposal)

- Corpus rebuild with a manifest: correct ESA source, public ISRO material, additional NASA/ESA/JAXA reports.
- Structure-aware parsing with page provenance. Evidence-anchored causal extraction against a fixed schema.
- A human-annotated benchmark: causal factors, cross-mission analogy relevance, a QA set with categories.
- Hybrid and graph retrieval; baselines B0–B5; ablations; significance testing; judge validation.

5. Out of current scope (future research)

- Full multi-agency balance, especially ISRO, while key reports are not public.
- Fine-tuning a domain extractor or encoder (needs more labelled data than one annotator can produce).
- Formal causal models (STAMP/CAST control structures, fault trees) generated automatically.
- Real-time use on live anomaly telemetry; multimodal evidence (figures, tables, telemetry plots).
- Expert user study with practising mishap investigators.

11. Reviewer-level evaluation

Submitted today, the work would be rejected. The decisive reasons would be the absence of results, the small corpus, and novelty overlapping published KG-RAG work. Under the revised plan, the most dangerous remaining objection is small scale, and it can be answered only partly.

Likely reviewer objection	Severity	Response in the revised plan
"GraphRAG over failure reports is already published."	High	Narrow the claim to cross-mission causal analogy over long, heterogeneous spaceflight reports. Cite the prior work directly. The contribution is the evidence-anchored graph, the human-labelled benchmark and the comparative evidence
"25–35 reports is too few for general claims."	High	Unit of evaluation = questions (~150), passages (~600 labelled) and analogy pairs (all mission pairs). Report confidence intervals. Claim *within-corpus* findings only; position the corpus as a resource others can extend
"Your gold labels are one student's opinion."	High	Written annotation guideline; a second annotator on ≥25% of items; Cohen's κ / Krippendorff's α reported; disagreements adjudicated and logged
"The graph may simply add noise; show it helps."	High	Baselines B0–B5 plus ablations (no graph, no verification, no fields), broken down by question category. A null result is reported as such
"LLM-judge scores are unreliable."	Medium	Judge from a different model family than the generator; agreement with human ratings on a 25% sample; reference-based and claim-level metrics alongside
"Improvements are not significant."	Medium	Paired bootstrap / permutation tests, Holm correction, 3 generation seeds, effect sizes with 95% CIs
"Ground truth leaked into the system."	Medium	The schema and dictionary are frozen before test labels are seen; dev/test split by mission; test questions are written from reports, not from system output

Likely reviewer objection	Severity	Response in the revised plan
"Not reproducible with a closed LLM."	Medium	Open-weight models (≤8B, 4-bit) on one consumer GPU; released code, prompts, manifest and labels (subject to source licences)
"Agency balance is claimed but ISRO is missing."	Medium	State the limitation; do not claim cross-agency generality beyond the agencies actually represented

The eight questions you asked, answered as a reviewer would:

1. *Precise gap?* There is no evaluated method for retrieving causally analogous failures across long, unstructured, multi-agency spaceflight investigation reports with verifiable evidence.
2. *What is new?* Evidence-anchored causal-factor graph construction with source-span verification, combined with graph-augmented hybrid retrieval. Evaluated on a human-labelled cross-mission analogy and QA benchmark that does not yet exist. Each component is known; the combination and the evidence are new.
3. *Why this method?* Spaceflight reports lack fixed fields, so the base paper's field-wise indexing must be recovered from structure. Causal analogy is defined over shared causal factors, which a typed graph represents explicitly and plain vector similarity does not.
4. *Evidence it works?* Statistically significant gains over B3 (the base paper's method) on cross-mission questions, plus higher citation precision and extraction precision than unverified LLM extraction.
5. *Fair comparison?* All systems share the corpus, chunking, generator, context budget and prompt template; only retrieval and graph context vary.
6. *Convincing?* Only with CIs, significance tests, agreement statistics and per-category results. These are mandated in the plan.
7. *Main threats?* Small corpus, single-site annotation, OCR noise in old reports, possible LLM pre-training exposure to famous accidents (Challenger, Columbia), and judge bias.
8. *Why should IEEE care?* It offers a reusable resource and evidence for a safety-engineering use case where wrong, uncited answers are costly. It also answers a future-work direction stated in a 2026 IEEE paper. **This holds only if the experiments deliver. It is not a promise of acceptance.**

12. Decisions needed before Stage 1

These choices change the plan and cannot be settled from the code. Each has a recommended default.

- [ ] **Research framing.** Adopt "evidence-anchored cross-mission causal analogy retrieval" (recommended), or keep the broader "GraphRAG QA system" framing with weaker novelty.
- [ ] **Corpus scope.** Expand to 25–35 public reports across NASA, ESA, JAXA and the public ISRO statements (recommended), and drop the two ISRO annual reports from the failure corpus (keep them only as distractor documents for retrieval).
- [ ] **Second annotator.** Identify a peer or the supervisor to double-label at least 25% of items. Without this, the human-label contribution is weak.
- [ ] **Timeline.** Confirm the remaining time and the target venue deadline. Stage effort (proposal §19) assumes about 14–18 weeks part-time.
- [ ] **Compute.** Local open-weight models only (recommended for reproducibility), or permission to spend on an API model as an independent judge.
- [ ] **Code baseline.** Repair the current code in place (recommended: keep ingestion, rebuild extraction, retrieval and evaluation) versus a clean rewrite.

Sources

Opened in full: the arXiv, DOI and OpenAlex records. ESA, ISRO and The Wire facts come from search-result summaries and should be re-checked on the pages themselves: Kim & Kim, arXiv 2606.13249 · IEEE DOI 10.1109/CASE69030.2026.11704480 · IEEE DOI 10.1109/ICSC67292.2026.00066 · arXiv 2604.13101 · arXiv 2608.11277 · arXiv 2606.06003 · arXiv 2404.16130 · ESA Schiaparelli inquiry page · ISRO GSLV-F10 FAC statement · OpenAlex API (bibliographic records for every DOI cited).